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人工智能与大模型 (II)

Tabular Foundation Models Comparison

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Abstract

Tabular foundation models (TFMs), such as TabPFN [1] and TabICL [2], provide significant zero-shot and in-context prediction capabilities on structured tabular data, bypassing the need of dataset-specific training. This study benchmarks the performance of two TFMs against two established gradient-boosted baselines (XGBoost [3] and LightGBM [4]) across 12 diverse datasets selected from TALENT [5]. We conduct a comprehensive evaluation covering predictive performance (Accuracy, Balanced Accuracy, and Macro-F1), inference latency, computational resource consumption, and scalability regarding dataset size and dimensionality. Our results indicate that while TFMs consistently demonstrate superior predictive performance, this improvement comes at the expense of higher time and memory consumption. Furthermore, we analyze the decision-making reliability of these models through confidence calibration metrics, revealing that both TFMs and gradient-boosted baselines exhibit suboptimal calibration. Our findings provide insights into the trade-offs between predictive capability and operational efficiency in modern tabular modeling.

Contents

1 Introduction

Tabular data—structured records organized in rows and columns—constitutes one of the most prevalent data formats across finance, healthcare, manufacturing, and scientific research. The recent emergence of Tabular Foundation Models (TFMs) represents a fundamental shift in how tabular prediction tasks are approached. Unlike conventional methods that require training a model from scratch for each dataset, TFMs such as TabPFN and TabICL are pre-trained on massive collections of synthetic or real tabular tasks and perform classification by conditioning on labeled context examples at inference time. This pre-training and in-context learning paradigm enables strong zero-shot prediction capabilities without dataset-specific training, challenging the long-standing dominance of gradient-boosted decision trees—particularly XGBoost and LightGBM—which have served as the established baselines for tabular prediction for decades.

While initial studies demonstrate that TFMs can achieve competitive or superior performance on benchmark datasets, there remains a need to systematically compare their behaviors with traditional tree-based methods across multiple dimensions. Existing evaluations often report aggregate predictive accuracy without jointly examining inference cost, memory consumption, scalability with respect to data properties, or decision-making reliability through confidence calibration—factors that critically affect practical deployability.

In this work, we systematically compare TFMs and GBDTs along four dimensions—predictive performance, computational cost, scalability, and confidence calibration—formalized as research questions in Section 2.

Our results indicate that while TFMs consistently achieve superior predictive performance, this improvement comes at the expense of substantially higher time and memory consumption. Furthermore, we find that all models exhibit suboptimal confidence calibration, revealing a shared limitation in decision-making reliability. These findings provide insights into the trade-offs between predictive capability and operational efficiency in modern tabular modeling.

2 Problem Definition

This study asks: *do TFMs definitively outperform GBDTs?* We decompose this into four research questions:

1. **Predictive performance.** Do TFMs achieve superior accuracy across diverse classification tasks, or is their advantage confined to specific regimes?

2. **Computational cost.** Does the inference time and memory overhead of TFMs negate any accuracy gains in practical deployment?

3. **Scalability.** How do performance and cost scale with increasing sample size and feature dimensionality under controlled conditions?

4. **Reliability.** Do both model families suffer from poor confidence calibration, assigning high confidence to incorrect predictions?

We address these questions through a systematic, controlled-variable benchmark comparing two TFMs (TabPFN, TabICL) and two GBDTs (XGBoost, LightGBM) on 12 classification datasets from TALENT, evaluated under identical train/test splits and CPU-only execution.

3 Related Works

3.1 Gradient-Boosted Trees

Gradient-boosted decision trees (GBDTs) have been the dominant approach for tabular data prediction for nearly a decade. XGBoost introduced a scalable tree boosting system with regularization, efficient handling of sparse data, and out-of-core computation, establishing itself as a go-to method in both Kaggle competitions and industrial applications. LightGBM further advanced this line of work by introducing gradient-based one-side sampling and exclusive feature bundling, achieving faster training speeds and lower memory usage without sacrificing predictive accuracy. Both methods handle heterogeneous feature types natively, require minimal preprocessing, and are robust to missing values and unscaled numerical features.

Despite their practical success, GBDTs are fundamentally dataset-specific: each new task requires retraining from scratch, and their performance depends heavily on hyperparameter tuning. They also do not benefit from knowledge learned on other datasets, which limits their effectiveness on small-data regimes where training signals are scarce.

3.2 Tabular Foundation Models

The concept of a foundation model for tabular data was pioneered by TabPFN, a Transformer-based model pre-trained on millions of synthetic classification tasks generated from a prior over randomly initialized neural networks. By framing tabular prediction as an in-context learning problem, TabPFN can produce predictions for unseen datasets without any parameter updates, conditioning solely on a set of labeled context examples.

TabPFN v2 scaled this approach with larger models, richer synthetic data generation priors, and improved context handling, demonstrating competitive or superior performance against tuned GBDTs on small-to-medium benchmarks. TabICL extended the in-context learning paradigm to larger datasets by redesigning the context encoding mechanism to handle thousands of samples and features more efficiently, addressing the scalability bottleneck that limited earlier models. Subsequent works, including TabDPT[6] and TabICLv2[7], have further explored directions such as real-data pre-training, retrieval-augmented context selection, and improved training strategies. Collectively, these models represent a shift from per-dataset training toward general-purpose tabular predictors that transfer knowledge across tasks.

Despite their rapid development, systematic comparisons between TFMs and traditional tree baselines remain limited in scope. Most original papers report accuracy on standard benchmarks but devote less attention to inference cost, memory footprint, or scalability behavior under controlled variation of data properties—dimensions that this work explicitly investigates.

3.3 Benchmarks for Tabular Data

Several benchmark platforms have been developed to standardize the evaluation of tabular machine learning methods. TALENT provides a unified toolbox encompassing hundreds of classification and regression datasets, along with implementations of numerous deep learning and tree-based models. It organizes datasets by task type, sample size, and feature composition, making it a practical resource for controlled experiments. TabArena [8] offers a continuously updated leaderboard that emphasizes fair comparison across model families, including standardized splits, consistent metric reporting, and computational cost analysis.

However, existing benchmarks aggregate results across heterogeneous datasets without disentangling the effects of individual data properties, making it difficult to attribute differences to specific data characteristics.

4 Algorithm Design

4.1 Controlled-Variable Evaluation Strategy

To isolate how model behavior depends on individual data properties, we adopt two orthogonal experimental axes: **sample-size scalability** varies the number of rows ($\sim$20 features held constant) to assess how cost and accuracy grow with dataset size, while **feature-count scalability** varies feature dimensionality (2,000 rows held constant) to assess sensitivity to input width. This design enables separate attribution of performance differences to row count versus column count, rather than conflating them in aggregate benchmarks.

4.2 Scalability Analysis

We characterize how computational cost scales with data dimensions by fitting a log-linear model:

$$\log(T) = \alpha \cdot \log(N) + C \tag{1}$$

where T denotes predict time or wall time, and N denotes the number of training samples or the number of features. The exponent α quantifies the growth rate: $\alpha \approx 1$ indicates approximately linear scaling, while $\alpha \approx 2$ indicates quadratic behavior. A larger α implies faster cost growth and worse practical scalability.

4.3 Confidence Calibration

Following the calibration evaluation framework of Guo et al.[9], we assess whether predicted class probabilities reflect true correctness likelihood. For each model, we compute four quantities from the predicted probabilities:

- **Mean Confidence:** the average confidence across all test predictions.
- **Correct Mean Confidence:** mean confidence on correctly predicted samples.
- **Wrong Mean Confidence:** mean confidence on incorrectly predicted samples.
- **Confidence Gap:** the difference between correct and wrong mean confidence. A larger gap indicates that the model better distinguishes when it is likely correct versus incorrect.

5 Experiment Design

5.1 Datasets

We select 12 classification datasets from the TALENT benchmark, organized along two controlled experiment lines detailed below.

Sample-size scalability evaluates model behavior as the number of rows increases while the number of features is held approximately constant (20~ 23). We select six binary classification datasets partitioned into three scale groups:

Table 1: Sample-Size Scalability Datasets

Scale Group	Dataset	Samples	Features	Classes
A (1k-3k)	pc1	1,109	21	2
A (1k-3k)	kc1	2,109	21	2
B (3k-10k)	sylvine	5,124	20	2
B (3k-10k)	ringnorm	7,400	20	2
C (10k-30k)	jm1	10,885	21	2
C (10k-30k)	default_of_credit_card_clients	30,000	23	2

Feature-count scalability evaluates model behavior as feature dimensionality increases while sample size is held constant at 2,000 rows. We use six mfeat datasets, all of which are 10-class digit recognition tasks:

Table 2: Feature-Size Scalability Datasets

Scale Group	Dataset	Samples	Features	Classes
F1(6-20)	mfeat-morphological	2,000	6	10
F2(20-100)	mfeat-zernike	2,000	47	10
F2(20-100)	mfeat-karhunen	2,000	64	10
F2(20-100)	mfeat-fourier	2,000	76	10
F3(100-300)	mfeat-factors	2,000	216	10
F3(100-300)	mfeat-pixel	2,000	240	10

5.2 Models

We compare four models spanning two paradigms:

TabPFN v2 is a Transformer-based tabular foundation model pre-trained on millions of synthetic classification tasks, performing zero-shot in-context prediction without any parameter updates. We use the *tabpfn* package with default settings and no fine-tuning.

TabICL extends the in-context learning paradigm to larger datasets through a redesigned context encoding mechanism. We use the *tabicl* package in its default zero-shot configuration without fine-tuning.

LightGBM is a gradient-boosted decision tree framework that employs gradient-based one-side sampling and exclusive feature bundling. We use the *lightgbm* package with default hyperparameters.

XGBoost is a widely adopted gradient boosting library with regularization. We use the *xgboost* package with default hyperparameters.

All models are used with fixed random seed 42. No hyperparameter tuning is performed for any model—tree baselines use their library defaults, and foundation models operate in zero-shot mode—to ensure a fair comparison of out-of-the-box behavior.

5.3 Evaluation Metrics

We evaluate all models along two dimensions:

1. **Predictive Performance.** Assessed through three complementary metrics:
    - **Accuracy:** the overall proportion of correctly classified samples.
    - **Balanced Accuracy:** the average per-class recall, which accounts for class imbalance.
    - **Macro-F1:** the unweighted mean of per-class F1 scores, suitable for multi-class evaluation.
2. **Computational Cost.** Captured by five timing and memory metrics:
    - **Fit Time:** time spent on model training or context preparation.
    - **Predict Time:** pure inference time on the test set.
    - **Wall Time:** total elapsed time from experiment start to result file completion.
    - **Seconds per Test Sample:** predict time divided by the number of test samples.
    - **Peak Memory:** maximum Resident Set Size (RSS) of the Python process during the experiment, sampled every 0.05–0.10 seconds using `psutil`.

5.4 Hardware

Experiments are conducted on two device classes to assess whether timing and memory conclusions are hardware-dependent:

- **Light laptop:** standard ultrabook-class machines with integrated graphics, used by two team members. Representative of resource-constrained CPU-only environments.
- **Gaming laptop:** machines with discrete GPUs and higher-power CPUs, used by three team members. Representative of more capable consumer hardware.

All models run exclusively on CPU. Accuracy metrics are deterministic given the fixed splits and seeds, so they are reported across all five runs. Time and memory results are reported separately by device class, since these measurements depend on hardware.

6 Results and Analysis

6.1 Overall Predictive Performance

We first examine which models achieve the strongest predictive performance across all 12 datasets. Table 3 summarizes the mean Accuracy and Macro-F1 per model.

Table 3: Predictive Performance Comparison

Model	Mean Accuracy	Mean Macro-F1
TabICL	0.913	0.851
TabPFN	0.906	0.844
XGBoost	0.883	0.825
LightGBM	0.883	0.824

Both TabICL and TabPFN outperform baselines on average, with TabICL holding a consistent edge over TabPFN. However, Figure 1 shows that the advantage of TFMs is not uniform across datasets. On strongly separable binary tasks such as `sylvine` and `ringnorm`, all four models exceed 0.94 accuracy and the gaps between them are negligible. The advantage of TFMs is most pronounced in the mfeat series, particularly the mid-dimensional variants: TabICL improves Macro-F1 by 11.6 pp on zernike and 5.0 pp on fourier, though on karhunen the gap narrows to 2.8 pp. On the largest binary datasets (`jm1`, `default_of_credit_card_clients`), all models struggle similarly, with accuracy clustering tightly around 0.81~ 0.83 and Balanced Accuracy below 0.67, reflecting heavy class imbalance rather than model weakness.

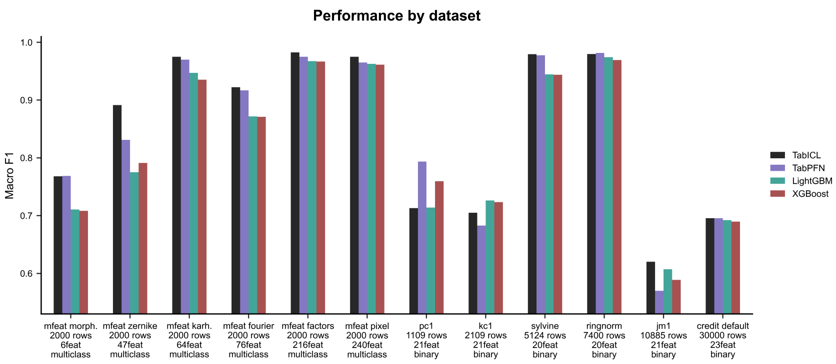


Figure 1: Performance by Dataset

In summary, TFMs deliver better average predictive performance, but their advantage is concentrated on medium-complexity multi-class tasks. On easy or severely imbalanced binary tasks, tree baselines are nearly indistinguishable.

6.2 Sample-Size Scalability

To understand how model behavior evolves with data volume, we compare performance and cost across the six sample-scale datasets, ranging from 1,109 to 30,000 training samples.

Time cost: The more consequential divergence emerges in time cost. As sample size increases from 1k to 30k, tree baselines remain below 30 ms throughout. TabICL predict time rises from 3.7 s to 419 s (mean across three gaming laptops; range 2.0~ 5.5 s to 230~ 598 s), and TabPFN from 14.4 s to 3,551 s (range 7.1~ 21.3 s to 1,716~ 4,572 s). This is captured quantitatively by the scalability exponent α, estimated via log-log linear regression (Figure 2):

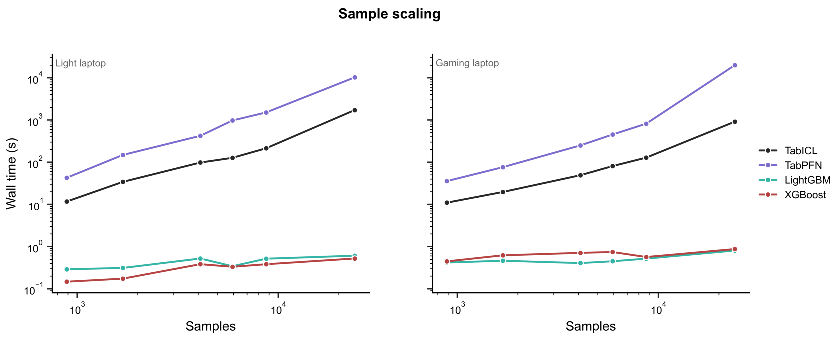


Figure 2: Time Cost Sample Scaling

Table 4: α Comparison on Sample Sizes

Model	α (predict time, Gaming)	α (predict time, Light)
LightGBM	0.45	0.70
XGBoost	0.37	0.30
TabICL	1.41	1.50
TabPFN	1.64	1.65

Table 4 reports the fitted α values. An exponent $\alpha \approx 1.0$ indicates approximately linear growth; whereas $\alpha \approx 2.0$ suggests quadratic behavior. TabPFN's scaling factor($\alpha \approx 1.6$) reflects substantially super-linear time growth relative to sample size; specifically, doubling the number of samples increases the prediction time by a factor of approximately $2^{1.6} \approx 3.03$. TabICL scales slightly better at $\alpha \approx 1.4 \sim 1.5$. However, both foundation models scale significantly worse than the decision tree baselines($\alpha \approx 0.3 \sim 0.7$). Specifically, LightGBM and XGBoost maintain sub-linear growth with respect to sample size in prediction runtime. Figure 3 further shows the per-device breakdown of these trends.

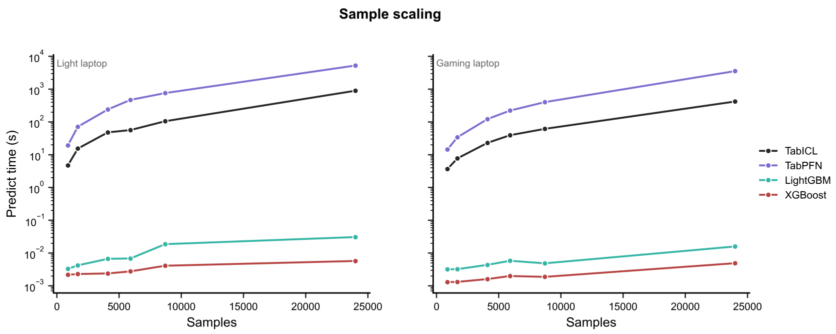


Figure 3: Sample Scaling

Memory cost: Peak memory follows a similar pattern. Tree baselines remain nearly flat at 150~ 250 MB regardless of sample size, while TabICL grows from ≈ 1.4 GB on `pc1` to ≈ 18.8 GB on `default_of_credit_card_clients` (gaming laptop), and TabPFN from 1.0 GB to 4.8 GB. The α values for memory growth confirm this: TabICL $\alpha \approx 0.76 \sim 0.84$, TabPFN $\alpha \approx 0.43$, while tree models have $\alpha \approx 0.02 \sim 0.04$ (Figure 4).

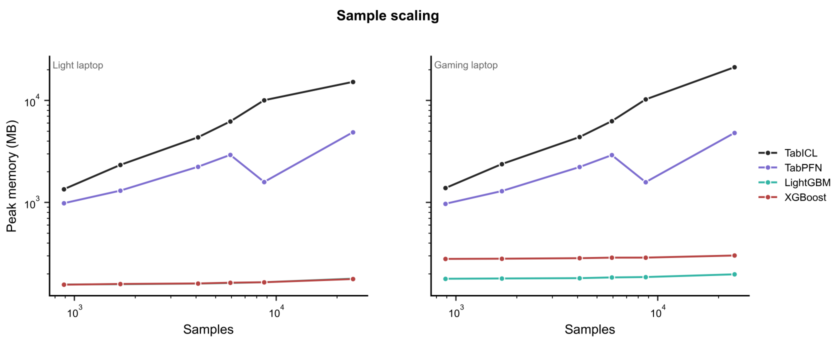


Figure 4: Memory Sample Scaling

Therefore, although TFMs maintain a predictive advantage across increasing sample sizes, this advantage comes at the cost of rapidly growing inference time and memory, which may limit their practicality on larger CPU-based tabular datasets.

6.3 Feature-Count Scalability

We now examine the second controlled-variable line: six mfeat datasets with identical sample size (2,000 rows) but varying feature dimensionality from 6 to 240.

Time cost: We quantify how predict time scales with feature count via the log-log exponent α (Table 5).

Table 5: α Comparison on Features

Model	α (predict time, Gaming)	α (predict time, Light)
LightGBM	0.07	-0.27
XGBoost	-0.01	-0.09
TabICL	0.68	0.69
TabPFN	0.94	0.82

Tree baselines show near-zero or negative α, meaning their inference time is largely independent of feature dimensionality. TFMs, by contrast, scale substantially with feature count: doubling the number of features roughly doubles predict time for TabICL ($\alpha \approx 0.69$) and slightly more than doubles for TabPFN ($\alpha \approx 0.82 \sim 0.94$). On `mfeat-pixel` (240 features, 2,000 samples), TabPFN predict time reaches 293 seconds on a gaming laptop versus 0.02 seconds for LightGBM—a four-orders-of-magnitude gap (Figure 5).

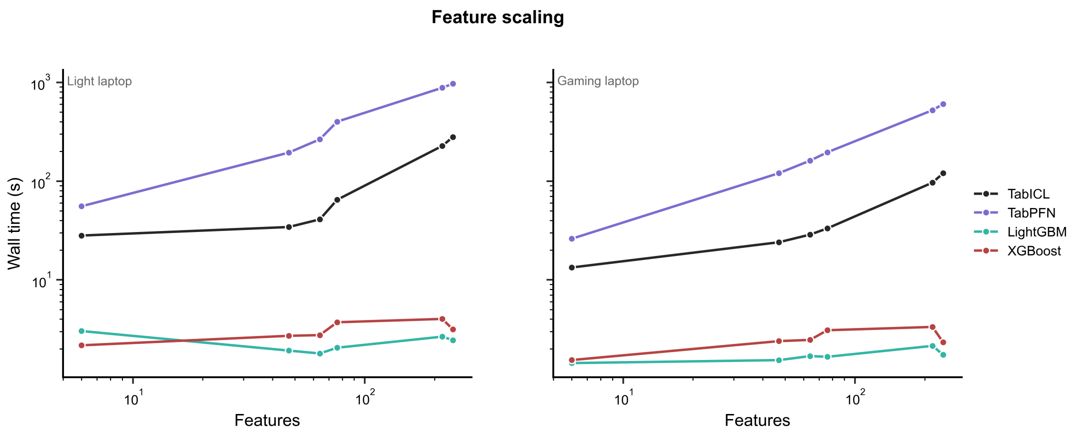


Figure 5: Feature Scale Loglog Time

Memory cost: Memory scaling with feature count is less extreme than with sample size (Figure 6). TabICL memory grows from 1.2 GB (6 features) to 17.5 GB (240 features), while tree baselines remain at 190–250 MB throughout. TabPFN memory grows from 0.7 GB to 2.2 GB.

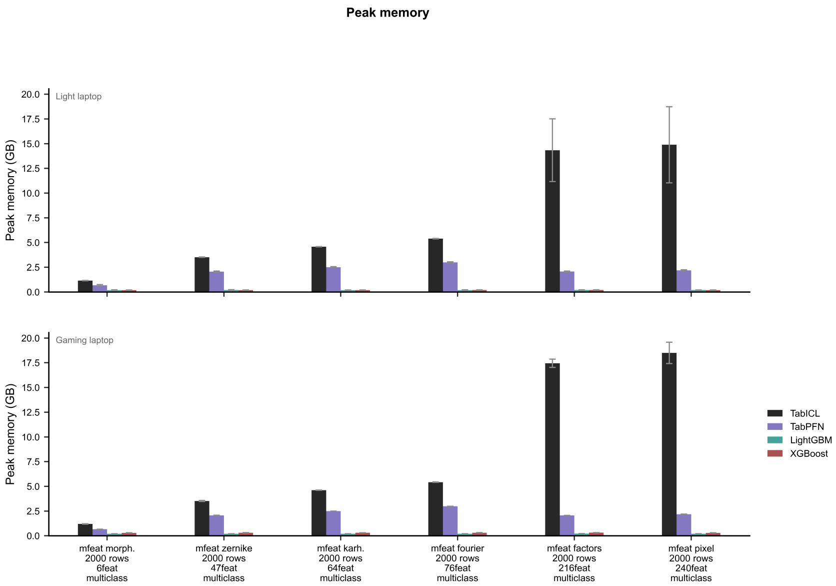


Figure 6: Feature Scale Peak-Memory Bar by Device

Feature dimensionality has a modest impact on tree models but a substantial one on TFMs, particularly TabPFN. However, the absolute costs are far lower than for sample-size scaling, even at 240 features, the 2,000-sample `mfeat-pixel` runs in under 5 minutes for TabPFN, whereas 30,000 samples $\times$ 23 features requires over an hour. This suggests that feature count and sample count interact in determining TFMs cost, but sample count dominates.

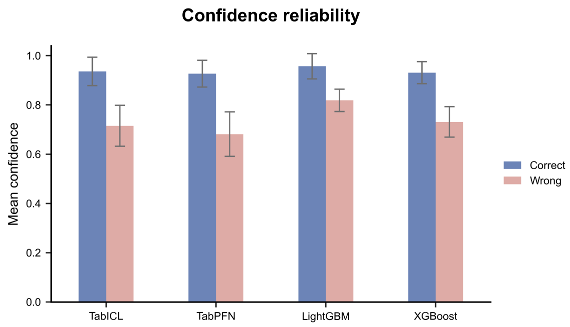
6.4 Confidence Calibration

Beyond predictive accuracy, we examine whether models can assess the reliability of their own predictions through confidence calibration[9]. Figure 7 compares the mean confidence assigned to correct versus incorrect predictions for each model.

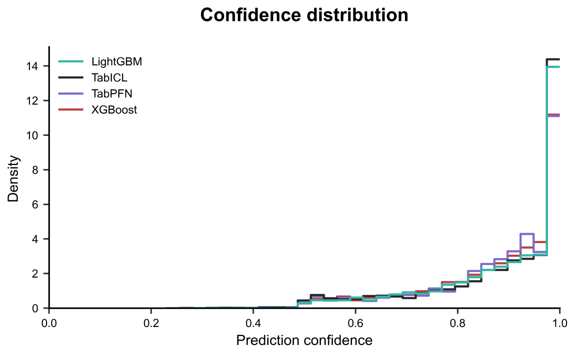
All four models assign substantially higher confidence to correct predictions than to incorrect ones, confirming that confidence carries signal about prediction quality. However, the degree of separation varies. TabICL and TabPFN exhibit the largest confidence gaps, exceeding 0.30 on several mfeat datasets—meaning that correct predictions are assigned confidence near 0.92~0.99 while wrong predictions receive around 0.54~0.65. In contrast, LightGBM and XGBoost produce narrower gaps: on `mfeat-morphological`, LightGBM's confidence gap is 0.05 (correct mean 0.95 vs. wrong mean 0.90), meaning the model remains highly confident even when mistaken.

A concerning pattern emerges across all models: wrong predictions routinely receive non-trivial confidence. On the binary dataset `jm1`, LightGBM assigns a mean confidence of 0.76 to its errors, and TabPFN assigns 0.76. Even TabICL, which shows the best overall separation, assigns 0.76 to wrong predictions on the same dataset. This indicates that none of the models provide a reliable "I don't know" signal—incorrect predictions are not accompanied by low confidence.

Confidence scores serve as a directionally useful signal for all models, yet the absolute calibration is unreliable. Users should not interpret high confidence as a guarantee of correctness, regardless of the model family.



(a) Confidence Reliability for Correct and Wrong Predictions across Models



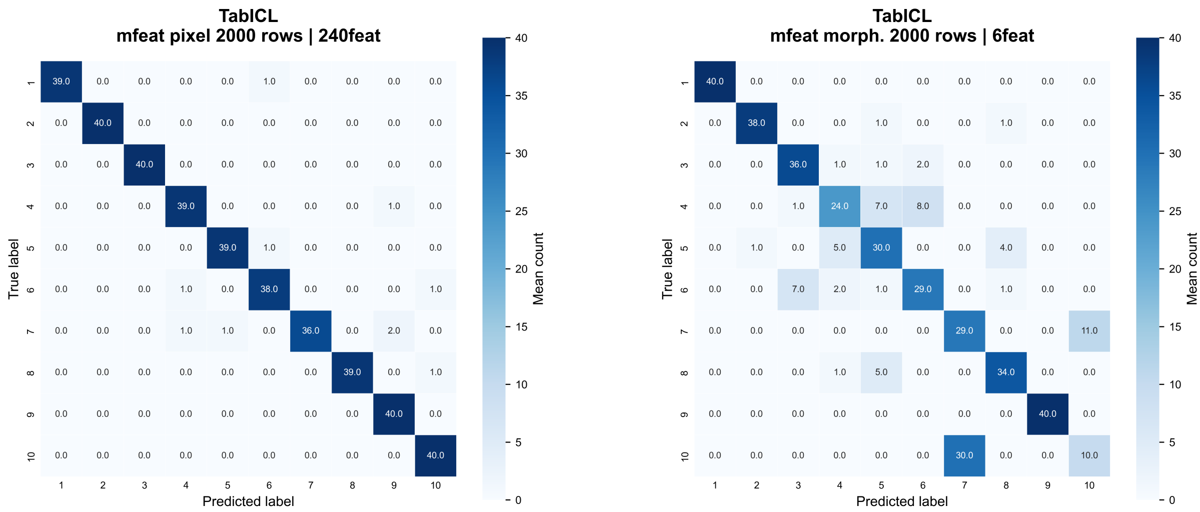
(b) Confidence Histogram by Model

Figure 7: Analysis of model prediction confidence across TFMs and GBDTs

6.5 Error Patterns

For the multi-class mfeat datasets, Figure 8 visualizes two representative confusion matrices and Table 6 lists the most frequent off-diagonal confusions across models. The digit pair 7~10 dominates errors on five of six datasets across all four models. On `mfeat-morphological` (6 features), LightGBM mispredicts true label 7 as 10 in 30 samples and 10 as 7 in 26—roughly half of all errors—while TabICL and TabPFN show the same pattern at comparable rates. This model-agnostic consistency indicates the error stems from feature ambiguity rather than model weakness: with only 6 morphological features, digits 7 and 10 are indistinguishable in feature space.

Higher dimensionality largely resolves these confusions: on `mfeat-pixel` (240 features), 7~10 errors drop to 1~3 samples per model with accuracy above 0.96. Other recurring confusions (4~5, 4~6, 6~4) similarly diminish as feature count increases.



(a) mfeat-pixel (240 features)

(b) mfeat-morphological (6 features)

Figure 8: Confusion Matrixes of TabICL

Table 6: Most Frequent Off-Diagonal Confusions in Multi-Class Mfeat Datasets

Dataset (Feat.)	Confusion	LightGBM	TabICL	TabPFN	XGBoost
morph. (6)	7 → 10	30	11	12	28
	10 → 7	26	30	29	26
	4 → 5	8	7	7	9
zernike (47)	10 → 7	30	15	29	27
	7 → 10	24	12	21	23
fourier (76)	10 → 7	17	10	10	17
	7 → 10	13	8	8	11
pixel (240)	8 → 10	3	1	2	2

7 Conclusion

This study compares tabular foundation models (TFMs) against gradient-boosted decision trees (GBDTs) across three critical dimensions: predictive performance, computational consumption, and reliability.

Our findings reveal a fundamental trade-off. TFMs achieve modest accuracy gains over GBDTs on medium-complexity multi-class tasks, but this advantage vanishes on simple binary or severely imbalanced datasets. More critically, the performance gains come at prohibitive computational cost: TFMs require orders of magnitude more inference time and memory than GBDTs, with super-linear scaling in sample size that renders them impractical for large-scale deployment. On reliability, both paradigms exhibit poor confidence calibration—incorrect predictions frequently carry high confidence, posing risks for high-stakes applications.

These results underscore that GBDTs remain the pragmatic choice for most real-world tabular tasks, offering robust accuracy, exceptional efficiency, and predictable scaling. TFMs may find niche utility where marginal accuracy improvements justify extreme resource expenditure, but their operational overhead and reliability limitations currently outweigh their benefits for general use.

Limitations include the focus on classification without hyperparameter tuning. Future work will extend this analysis to regression tasks, missing-value robustness, and GPU-accelerated inference scenarios.

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